

# The Accuracy Illusion: How Overlapping Samples Manufacture Confidence, Not Skill, in Intraday Index Forecasting

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**Abstract**—Intraday forecasting systems routinely report directional hit-rates between 60 % and 85 %, and those figures move real capital. This paper shows that they are not valid estimates of predictive skill, and measures precisely how they arise. When a  $k$ -bar-ahead forecast is issued every bar, consecutive forecasts share  $k-1$  of their  $k$  outcome bars. If the model’s own calls also persist, and on a live production system we measure a mean run of 15.2 minutes, the reported hit-rate acquires a variance far larger than its nominal sample size implies. We simulate a forecaster with *exactly zero skill* against real NSE index bars. It posts a  $\geq 60$  % session once in six and a  $\geq 71$  % session once in 51, while reporting a confidence interval  $1.66\times$  too narrow. Across a panel of two indices and four horizons, a  $\geq 71$  % reading surfaces every seventh session. Under correct per-event scoring, nine model families spanning logistic regression to a Temporal Fusion Transformer, all trained on identical purged walk-forward splits, reach at most 51.3 % out-of-sample when scored per bar, and *none* remains significant once scored per event or at its effective sample size; a model-independent ceiling test replicates across five indices and two ex-changes. A sweep of 86 rule-based strategies yields a best Sharpe *below* what a circular block-bootstrap null produces from selection alone ( $P = 0.658$ ). We give the closed form behind the distortion, showing the variance inflation is governed by the second moment of the run-length distribution, and an estimator built on it lifts the coverage of a nominal 95 % interval from 40 % to 94 %; the corrected effective sample size of a 69-forecast session is six. We then quantify what would actually be required. Break-even accuracy derived from measured move distributions is 71.5 % at the 30-minute horizon, and it exceeds 100 % at shorter horizons and at option-level costs. The correction is applied to our own results and removes the one positive directional effect we had found: XGBoost falls from 51.3 % ( $p = 0.001$ ) to 50.4 % ( $p = 0.69$ ). Applied to a volatility target on identical features it removes nothing (70.8 % to 71.3 %), which is evidence that the correction separates signal from artefact rather than flattening both. All code, raw results and the free-data corpus are public.

**Index Terms**—overlapping observations, backtest overfitting, directional forecasting, market microstructure, multiple hypothesis testing, negative results, reproducibility, index derivatives

## I. INTRODUCTION

A forecasting system that reports 71 % directional accuracy is, on its face, extraordinary. Work that controls for scoring unit, transaction costs and multiple testing places realistic out-of-sample directional accuracy for liquid index intraday prediction close to the coinflip [13], [21], while surveys of the applied literature catalogue a great many higher figures reported without those controls [32]. Yet such numbers appear regularly on trading dashboards, in vendor marketing and in applied publications, and they are frequently computed in good faith from data the system itself generated.

This paper began as an attempt to build such a system. Over fourteen months we built and ran a live multi-horizon forecasting platform for Indian index derivatives, and it reported a headline accuracy of 71 %. That number was wrong. Tracing it to source revealed no bug in the model. The defect lay in the *denominator*, in how a per-minute forecast stream is scored. Four further positive results from the same project were later retracted for related reasons.

The econometrics of overlapping observations is not new [1], [2]. Hansen and Hodrick [1] established four decades ago that sampling data more finely than the forecast interval induces serial correlation in forecast errors and invalidates naive standard errors. What that literature does not document, and what we supply here, is the *magnitude and shape* of the distortion in a now-ubiquitous practice: the every-minute multi-horizon directional dashboard. There the overlap meets a second ingredient, prediction persistence, and the product is then amplified by the multiplicity of an index  $\times$  horizon panel.

### A. Contributions

- 1) **A measurement of the artefact as it occurs in practice.** Overlap on its own changes nothing, since a forecaster that re-draws every bar yields an i.i.d. hit indicator. Inflation requires overlap *and* persistent predictions together. We calibrate persistence from a production system’s own forecast log, then

quantify the resulting distribution, its calibration error and its panel multiplicity (Section V).

- 2) **A complete control set for a negative result.** Shuffled-label controls on every model, injected synthetic signals of known strength as a positive control, nine architectures under identical purged splits, and a block-bootstrap null for the strategy sweep (Section VII). The positive control is essential. Without it, “we found nothing” cannot be distinguished from “our pipeline is broken.”
- 3) **A closed form and a corrected estimator.** We derive the effective sample size of an overlapping, persistent forecast stream and show it is governed by  $\mathbb{E}[L^2]/\mathbb{E}[L]$  over the run lengths. An interval built on it covers at 94 % where the naive interval covers at 40 % and a standard HAC correction at 67 % (Section VI).
- 4) **The required-versus-achieved framing.** We derive break-even accuracy from measured move distributions and show that below a horizon threshold *no* accuracy is sufficient, which reframes high-accuracy claims as either artefacts or as far less valuable than they appear (Section VIII).
- 5) **A reusable free-data corpus** for NSE index derivatives, including a decade of end-of-day option chains and two years of participant-wise open interest, with the observation that the exchange’s static archive is not geo-restricted while its live API is (Section III).

All experiments are reproducible from public code and committed raw results.<sup>1</sup>

## II. RELATED WORK

### A. Overlapping observations

Hansen and Hodrick [1] and Richardson and Smith [2] characterise inference under overlapping returns; Newey and West [6] give the standard heteroskedasticity- and autocorrelation-consistent correction. Boudoukh, Richardson and Whitelaw [3] are the closest antecedent to this paper: they show that overlapping observations manufacture the appearance of long-horizon return predictability, and that the apparent effect largely disappears under correct inference. Our subject is the same distortion at the opposite end of the horizon scale, and applied to a different reported quantity.

That literature concerns regression coefficients and their standard errors. We apply the same logic to what practitioners actually publish, the session directional hit-rate, and measure the consequence empirically rather than asymptotically. The correction we arrive at is the classical design effect [4], [5] with clusters given by runs of identical calls; the contribution is its application to forecast streams, not the statistic itself.

### B. Backtest overfitting and multiple testing

Harvey, Liu and Zhu [15] argue that the conventional  $t > 2$  threshold is indefensible after decades of factor mining and propose  $t > 3$ . Bailey and López de Prado [16] introduce the Deflated Sharpe Ratio, and Bailey *et al.* [17] the Probability of Backtest Overfitting via combinatorially symmetric cross-validation. White’s Reality Check [12], its refinement by Hansen [14] and the Benjamini–Hochberg procedure [11] provide complementary family-wise and false-discovery-rate controls. Sullivan, Timmermann and White [13] apply the Reality Check to a large universe of technical trading rules, which is the direct precedent for our strategy sweep; Section VII-C is best read as a replication of that design on intraday index data with a modern cost model. We apply these corrections and report openly where each one fails on our data and why.

### C. Machine learning for financial prediction

Fischer and Krauss [30] apply LSTM networks to S&P 500 constituents, reporting daily returns of 0.46 % and a Sharpe ratio of 5.8 *before transaction costs*; Krauss *et al.* [31] compare deep networks, gradient-boosted trees and random forests in the same setting. Gu, Kelly and Xiu [29] conduct a comprehensive machine-learning comparison for asset pricing, reporting modest out-of-sample  $R^2$ . Zhang *et al.* [28] achieve strong accuracy on limit-order-book data, and Kolm *et al.* [27] demonstrate genuine multi-horizon alpha from order-flow imbalance. Both results rest on message-level exchange data rather than on bars, which is the distinction that matters for our purposes. Sezer *et al.* [32] survey the field.

Table I compares this work with representative prior studies along the methodological axes that determine whether a reported accuracy is interpretable. We deliberately compare on *procedure* rather than on headline numbers: our claim is not that prior results are wrong, but that several widely used reporting practices admit the artefact documented here, and that a reported accuracy cannot be assessed without knowing the scoring unit and the cost model.

### D. What is predictable

Volatility clustering [22], [23] and the intraday volatility U-shape [24] are among the most robust regularities in financial time series, and they concern magnitude rather than direction. The variance risk premium, meaning the persistent gap between implied and subsequently realised volatility, is documented by Carr and Wu [25] and by Bakshi and Kapadia [26]. Our positive results fall entirely within these two categories, and we regard that as the paper’s substantive finding rather than as a consolation.

## III. DATA

All data is obtainable free of charge from a retail brokerage account and public exchange archives. Table II summarises the corpus.

<sup>1</sup><https://github.com/rachitranka25/accuracy-illusion>

TABLE I

METHODOLOGICAL COMPARISON WITH REPRESENTATIVE PRIOR WORK. “SCORING UNIT” IS THE UNIT IN WHICH HIT-RATE OR RETURN IS AGGREGATED; PER-BAR SCORING OF A MULTI-BAR HORIZON IS THE CONDITION UNDER WHICH THE ARTEFACT OF SECTION V APPEARS. MT: MULTIPLE-TESTING CORRECTION. NC: NULL CONTROL (SHUFFLED LABELS OR EQUIVALENT). C: TRANSACTION COSTS MODELLED. R: CODE AND RAW RESULTS PUBLIC.

| Study                           | Market      | Data granularity       | Scoring unit                          | MT      | NC | C         | R       |
|---------------------------------|-------------|------------------------|---------------------------------------|---------|----|-----------|---------|
| Fischer & Krauss [30]           | S&P 500     | Daily bars             | Per day, non-overlapping              | —       | —  | pre-cost  | —       |
| Krauss <i>et al.</i> [31]       | S&P 500     | Daily bars             | Per day, non-overlapping              | —       | —  | pre-cost  | —       |
| Gu <i>et al.</i> [29]           | US equities | Monthly                | Per month                             | partial | —  | partial   | partial |
| Zhang <i>et al.</i> [28]        | LSE         | Message-level LOB      | Per event                             | —       | —  | —         | partial |
| Kolm <i>et al.</i> [27]         | Nasdaq      | Message-level LOB      | Per horizon                           | —       | —  | discussed | —       |
| Applied index studies (typical) | NSE, global | Daily / intraday bars  | Often per bar                         | —       | —  | —         | rare    |
| <b>This work</b>                | NSE index   | 5-min bars, EOD chains | <b>Per event (<math>k</math>-bar)</b> | ✓       | ✓  | ✓         | ✓       |

This work applies Bonferroni, Benjamini–Hochberg, the Deflated Sharpe Ratio and PBO/CSCV (MT); shuffled-label controls on every model plus injected positive controls (NC); explicit round-trip costs, and inverts them into a required-accuracy threshold (C).

TABLE II  
FREE-DATA CORPUS. ALL COLLECTORS ARE PUBLIC.

| Asset                          | Period            | Size      |
|--------------------------------|-------------------|-----------|
| BANKNIFTY 5-min bars           | 2025-02 – 2026-07 | 26 265    |
| NIFTY 50 5-min bars            | 2025-02 – 2026-07 | 26 265    |
| FINNIFTY / MIDCPNIFTY / SENSEX | 2025 – 2026       | 9.6k–14k  |
| Option chain EOD + OI          | 2024-07 – 2026-07 | 1 382 738 |
| Long-history option chain      | 2016-05 – 2026-05 | 2 181 d   |
| Participant-wise OI            | 2024-07 – 2026-07 | 515 d     |
| Index futures (real volume)    | 2024-01 – 2026-07 | 639 d     |

Three observations about this corpus are of independent practical value.

**Index spot carries no volume.** The cash index prints zero volume, so any feature computed as a “VWAP” on index spot is in fact an unweighted expanding mean of typical price. This was live in our own feature set for weeks before it was identified. Real volume exists only on the futures contract.

**The static exchange archive is not geo-restricted.** The exchange’s *live* option-chain API returns HTTP 403/404 from outside India, but the static end-of-day derivatives archive downloads normally. This distinction converted an assumed hard constraint into a two-year, and subsequently ten-year, dataset.

**Time-based one-time-password authentication is location-independent.** Brokers authenticating by locally computed TOTP permit data collection from any jurisdiction, whereas SMS-based flows do not. This is a deciding practical constraint on reproducibility, and it is not documented prominently anywhere.

#### IV. METHODOLOGY

Every accuracy reported in this paper passes through one shared harness, so that differences between models are differences between models.

**Purged walk-forward.** Expanding-window splits, with the  $k$  bars whose labels overlap the test window removed from the end of each training set. Label overlap is the most common leakage channel in this setting [18].

**Exact intervals.** Accuracies are reported with Clopper–Pearson 95 % confidence intervals and the exact binomial  $p$ -value against 0.5. We additionally report the Pesaran–Timmermann statistic [10], which is the appropriate null for directional accuracy: it compares the realised hit-rate against what independence between forecast and outcome would produce given their own marginal rates, and therefore catches a model that merely tracks drift.

**Shuffled-label controls.** Every model is additionally trained and evaluated with randomly permuted labels. If real and shuffled scores coincide, the pipeline is reading noise. That is a stronger statement than any train/test gap, since it does not require the model to be well specified.

**Positive controls.** A synthetic feature agreeing with the label on a known fraction  $s$  of rows is injected, making the recoverable accuracy approximately  $0.5 + s/2$ . A pipeline that fails to recover it cannot be trusted to report a null.

**Per-event scoring and costs.** Strategy returns are evaluated on non-overlapping  $k$ -bar holds with explicit round-trip costs, and costs are treated as part of the hypothesis rather than as a subsequent adjustment.

**Units.** One conversion is used throughout and is stated here once to avoid ambiguity later. All analysis is on 5-minute bars, so the 30-minute horizon is  $k = 6$  bars and a session of 09:15–15:30 yields  $n = 69$  scored forecasts. The production system emits a forecast every *minute*, so its run lengths are measured in minutes and divided by five to obtain bars: a mean run of 15.2 minutes is 3.0 bars and the longest observed run of 205 minutes is 41 bars. Because the live stream is five times denser than our simulation while carrying the same underlying information, the nominal-to-effective ratio it reports is worse than the one we measure, and our figures are therefore conservative with respect to the artefact they describe.

Features are the fifteen columns used by the production system: one-bar return, RSI, EMA distance, MACD histogram, Bollinger width and %b, ATR%, Parkinson volatility, VWAP distance, time-of-day, day-of-week,

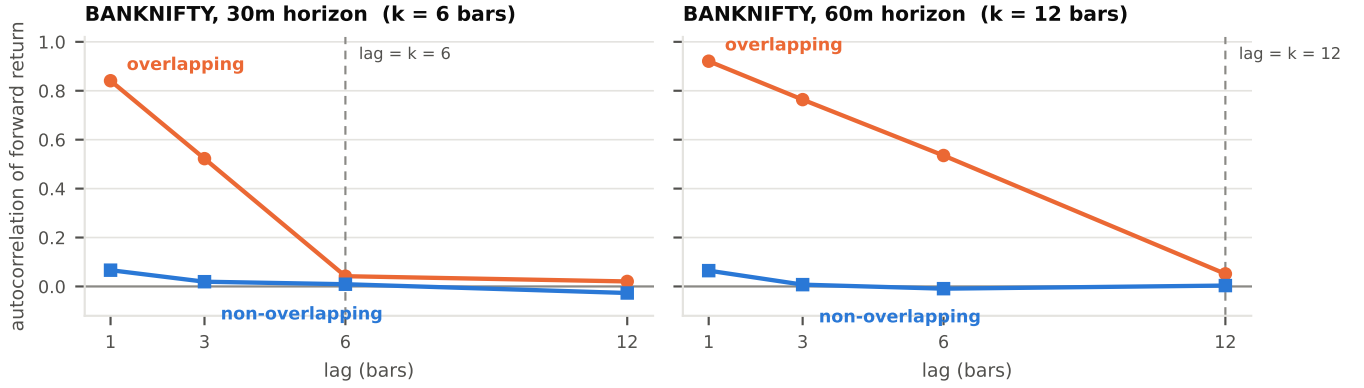


Fig. 1. Autocorrelation of the forward return, overlapping versus non-overlapping, at two horizons. In both panels the overlapping series decays to zero at exactly lag  $k$ , which is the signature of arithmetic overlap rather than of serial structure.

regime volatility, Hurst exponent, and distance from the previous session's high and low. All are transforms of a single price series, a point we return to in Section IX.

## V. THE MECHANISM

### A. Overlap is arithmetic, not information

Fig. 1 shows the autocorrelation of the  $k$ -bar forward return measured two ways. Computed on overlapping windows it appears strongly autocorrelated, at 0.841 for lag 1, and decays to approximately zero at *exactly* lag  $k$ . Sampled so that no two observations share a bar, the autocorrelation is 0.067 at lag 1 and indistinguishable from zero thereafter. The pattern reproduces independently at  $k = 6$  (30 minutes) and  $k = 12$  (60 minutes), with the decay landing on lag 6 and lag 12 respectively. The correlation is a property of the sampling scheme, not of the market.

### B. Persistence is the second, necessary ingredient

Overlap alone is insufficient. If predictions are drawn independently each bar, the hit indicator  $\mathbf{1}\{\hat{y}_t = y_t\}$  is i.i.d. Bernoulli regardless of how autocorrelated  $y_t$  is, and no inflation occurs. Our first simulation made precisely this error and correctly found nothing.

Real models do not re-decide every bar: their features move slowly, so they repeat a call for long stretches. Measured on the production system's own forecast log at the 30-minute horizon, BANKNIFTY calls persist for a mean of 15.2 minutes (99 runs across 1 500 forecasts, longest 135 minutes) and NIFTY 50 for 12.2 minutes (123 runs, longest 207 minutes). A model re-drawing each bar would show a mean run of one.

When both the call and the outcome persist, a single trending stretch is recorded as dozens of hits.

### C. What a zero-skill forecaster reports

We simulate a forecaster with no skill whatsoever: a fair coin, held for a run length drawn from the empirical distribution above. It is evaluated against real BANKNIFTY

TABLE III

A FORECASTER WITH EXACTLY ZERO SKILL, 30-MINUTE HORIZON, SCORED PER BAR VERSUS PER EVENT. BANKNIFTY; NIFTY 50 IS WITHIN 0.3 POINTS ON EVERY ROW.

|                                  | Per bar                        | Per $k$ bars                   |
|----------------------------------|--------------------------------|--------------------------------|
| Reported sample size per session | 69                             | 12                             |
| Stated binomial 95 % CI          | $\pm 11.8$ pts                 | $\pm 28.3$ pts                 |
| Realised spread across sessions  | $\pm 19.6$ pts                 | $\pm 28.0$ pts                 |
| <b>Understatement factor</b>     | <b>1.66<math>\times</math></b> | <b>0.99<math>\times</math></b> |
| $P(\text{session} \geq 60 \%)$   | 15.7 %                         | 19.2 %                         |
| $P(\text{session} \geq 71 \%)$   | 2.0 %                          | 7.1 %                          |
| $P(\text{session} \geq 85 \%)$   | $\approx 0.01 \%$ <sup>†</sup> | 0.30 %                         |

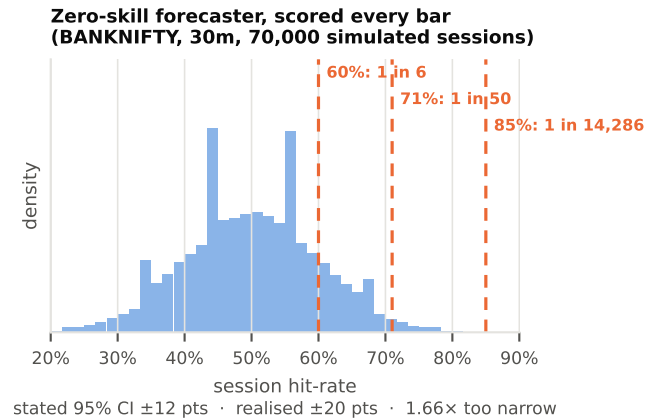


Fig. 2. Measured session hit-rate distribution of a zero-skill forecaster scored every bar. The multimodality is not noise: with roughly a dozen independent decisions per session the hit-rate lands on near-discrete values, which is the small effective sample made visible.

and NIFTY 50 bars and scored the way a live dashboard scores it. Results over approximately 70 000 simulated sessions per index appear in Table III and Fig. 2.

Table III makes a point that is easy to state backwards, so we state it carefully. Per-bar scoring does *not* produce

extreme session readings more often than honest scoring. It produces them *less* often: a  $\geq 71\%$  session appears on 2.0% of per-bar sessions against 7.1% when scored per event. What per-bar scoring does is report those readings with an interval  $1.66\times$  too narrow, because the quoted  $n$  of 69 is a fiction.

So overlap does not manufacture high hit-rates. It manufactures unwarranted *confidence* in them. A figure that should be read as “somewhere between incompetence and mastery” is instead read as a precise 71%, and it is that false precision, not the figure itself, that misleads. The calibration column is therefore the diagnostic throughout this paper, and the frequency columns are context.

#### D. Panel multiplicity

A dashboard does not monitor one number. The system studied here displays two indices at four horizons, so eight hit-rates every session. The two indices are strongly correlated intraday and the four horizons nest, so an independence calculation would be an upper bound of unknown tightness. We therefore simulate the whole panel jointly on the same sessions, each horizon carrying its own  $k$  and its own forecast stream, and count the joint event directly:

the per-series rate is 2.04%, an independence calculation gives 15.2%, and the measured panel gives **15.1%**. The approximation is adequate here, which is worth stating: although the outcome series are correlated, a zero-skill forecaster’s hits are close to independent across them. Either way the figure is about one session in seven, so a headline 71% arrives within the first weeks of live operation with no edge whatsoever.

Our own 71% was exactly this. Traced to source it was NIFTY 50, 30-minute horizon, on a single trending Friday: 202 of 285 per-minute forecasts scored as hits. The identical configuration scored 56% the following Monday, and 51% when re-evaluated out-of-sample at one observation per bar.

### VI. A CORRECTED ESTIMATOR

Diagnosing a biased statistic is only half a contribution. This section supplies the closed form behind the distortion and an estimator that removes it, then validates the estimator by coverage rather than by argument.

#### A. Effective sample size

Let  $h_t = \mathbf{1}\{\hat{y}_t = y_t\}$  denote the hit indicator of a forecaster issuing a  $k$ -bar-ahead call every bar, and let  $\hat{p} = n^{-1} \sum_t h_t$  be the reported hit-rate. Practitioners quote  $\text{Var}(\hat{p}) = p(1-p)/n$ , which holds only for independent  $h_t$ . In general,

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} \left[ 1 + 2 \sum_{j=1}^{n-1} \left( 1 - \frac{j}{n} \right) \rho_j \right], \quad (1)$$

where  $\rho_j = \text{Corr}(h_t, h_{t+j})$ . The bracketed term is the variance inflation factor, and dividing  $n$  by it gives the effective sample size  $n_{\text{eff}}$ .

TABLE IV  
COVERAGE OF A NOMINAL 95% INTERVAL ON A SESSION HIT-RATE, AGAINST A FORECASTER OF KNOWN SKILL WITH REALISTIC PERSISTENCE. BANKNIFTY, 30-MINUTE HORIZON, NOMINAL  $n = 69$ ; NIFTY 50 AGREES WITHIN 1 POINT THROUGHOUT.

| Estimator            | Coverage at skill |             |             | Width | $n_{\text{eff}}$ |
|----------------------|-------------------|-------------|-------------|-------|------------------|
|                      | 50 %              | 55 %        | 60 %        | (pts) |                  |
| Naive binomial       | 39.6              | 39.5        | 38.9        | 21.4  | 69               |
| HAC (Bartlett)       | 66.6              | 66.6        | 66.1        | 33.8  | 21               |
| Block bootstrap      | 63.1              | 63.0        | 63.0        | 29.6  | —                |
| <b>Run-structure</b> | <b>94.0</b>       | <b>94.1</b> | <b>94.2</b> | 70.8  | <b>5.6</b>       |

Two structures drive  $\rho_j$ , and the argument of Section V is that both are required. The outcome  $y_t$  is autocorrelated out to lag  $k$  because consecutive windows share bars, and the prediction  $\hat{y}_t$  persists in runs. If predictions are redrawn each bar,  $h_t$  is i.i.d. whatever  $y_t$  does, every  $\rho_j$  vanishes, and  $n_{\text{eff}} = n$ .

Because the dependence here is block-shaped, (1) admits a direct form. If the sequence decomposes into runs of length  $L_i$  within which  $h_t$  is constant, then conditioning on the lengths gives  $\text{Var}(\hat{p}) = p(1-p) \sum_i L_i^2/n^2$ , so

$$n_{\text{eff}} = \frac{n}{\sum_i L_i^2/n} \xrightarrow{n \rightarrow \infty} \frac{n \mathbb{E}[L]}{\mathbb{E}[L^2]}. \quad (2)$$

Equation (2) is the classical design effect [4], [5] with clusters given by the runs; what is new here is only the observation that a forecast stream has exactly this structure and that nobody reports it.

The inflation is governed by  $\mathbb{E}[L^2]$ , not  $\mathbb{E}[L]$ . A run-length distribution with a heavy tail inflates the variance far beyond what its mean suggests, and that is precisely the regime a real forecaster occupies: on our production log the mean run is 15.2 minutes while the longest is 135.

#### B. Validation by coverage

An interval is judged by whether it covers. We generate a forecaster whose true skill is fixed by construction, with calls persisting according to the measured run-length distribution, and ask how often a nominal 95% interval contains that known truth. Table IV reports four estimators.

Both HAC and the bootstrap improve substantially on the naive interval and both still under-cover. They fall short for the same reason, and it is a consequence of (2). The conventional automatic Bartlett bandwidth,  $4(n/100)^{2/9}$ , evaluates to 3.7 bars at  $n = 69$ , which is almost exactly the *mean* run length of 3.0 bars. But the variance inflation is driven by  $\mathbb{E}[L^2]$ , so it is the tail that matters, and the tail here reaches 41 bars. A kernel truncated near the mean has summed almost none of the covariance that the long runs contribute.

Estimating the inflation from the observed run structure removes that failure mode. It over-corrects in the other direction, because treating runs as independent clusters ignores that consecutive runs are negatively dependent, so

TABLE V

MODEL-INDEPENDENT CEILING TEST AT THE 30-MINUTE HORIZON, REPLICATED ACROSS FIVE INDICES. TRAINING ACCURACY IS 100.0 % ON EVERY ROW.  $\Delta$  IS THE ABSOLUTE GAP BETWEEN THE REAL AND SHUFFLED COLUMNS, COMPUTED BEFORE ROUNDING. THE POSITIVE CONTROLS HAVE RECOVERABLE ACCURACIES OF APPROXIMATELY 55 % AND 65 % BY CONSTRUCTION.

| Index      | $n$    | Real | Shuf. | $\Delta$ | Maj. | $s=.10$ | $s=.30$ |
|------------|--------|------|-------|----------|------|---------|---------|
| BANKNIFTY  | 16 856 | 50.5 | 50.2  | 0.3      | 49.3 | 52.4    | 63.5    |
| NIFTY 50   | 16 853 | 50.0 | 50.2  | 0.3      | 49.8 | 52.0    | 63.7    |
| FINNIFTY   | 8 988  | 50.2 | 50.5  | 0.3      | 49.4 | 52.4    | 62.5    |
| MIDCPNIFTY | 8 309  | 50.2 | 50.8  | 0.7      | 49.7 | 53.3    | 62.0    |
| SENSEX     | 6 144  | 49.7 | 50.1  | 0.5      | 49.0 | 52.5    | 62.1    |

No real-label result is significant against 0.5 ( $p = 0.22$  to  $0.91$ ); every positive control is, at  $p < 0.001$ .

we report it as a conservative bound rather than as an exact correction.

### C. What the corrected number says

The practical content is  $n_{\text{eff}}$  itself. A session that reports 69 forecasts carries roughly *six* independent observations, and an honest interval on its hit-rate is about  $\pm 35$  points wide. Such an interval spans everything from incompetence to apparent mastery, which is the correct description of what one session of a per-minute forecast stream can establish.

The correction changes no point estimate. Only the interval moves. What it buys is an interval that means what it claims, and the discipline of reporting  $n_{\text{eff}}$  alongside  $n$  whenever forecasts overlap.

## VII. PREDICTABILITY UNDER CORRECT SCORING

### A. Is there anything to predict?

We first ask the question in a form independent of any model. An unconstrained random forest was fitted until it memorised the training data, on five indices spanning large-cap, banking, financial-services, midcap and a second exchange (Table V).

The pattern is identical on all five. Each model memorises its training data completely and generalises at chance, and permuting the labels moves out-of-sample accuracy by between 0.3 and 0.7 points. No real-label result is distinguishable from a coinflip. Both positive controls are recovered on every index at  $p < 0.001$ , which establishes that the pipeline detects signal whenever signal is present and that the null is therefore a property of the data.

Replication across five instruments, two exchanges and four capitalisation segments is what separates this from a single-market curiosity.

### B. Nine model families

Table VI reports nine families on identical splits, features, seeds and controls. Fig. 3 plots the same results with exact intervals.

Scored the way a dashboard scores, three of eight BANKNIFTY models clear  $p < 0.05$  and XGBoost reaches

51.3 % at  $p = 0.001$ . That was, in an earlier draft of this paper, our one surviving positive directional result, and we described it as genuine and Bonferroni-robust.

It does not survive our own correction. Widening the interval to  $n_{\text{eff}}$  puts 50 % inside every interval in the table. Re-scoring one observation per 6 bars, so that no two observations share an outcome bar, moves XGBoost from 51.3 % ( $p = 0.001$ ) to **50.4 %** ( $p = 0.692$ ). Across both indices, the count of models significant at  $p < 0.05$  falls from four to **zero** under either correction.

We report this prominently because it is the paper’s own thesis applied to the paper’s own result, and because a correction that spares the author’s favourite finding would be worth very little. The directional null is not a near-null with one exception. It is a null.

Table VI also contains its own diagnostic. The NIFTY 50 Transformer sits at 48.3 % with  $p < 0.001$ , which at the nominal  $n$  is 4.4 standard errors *below* chance. Under a correctly specified null the largest  $|z|$  across eighteen tests should land near 2.6–2.9; observing 4.4 is not a bug in that model but the variance inflation of Section V surfacing inside our own results table. At  $n_{\text{eff}}$  its interval is  $[46.3, 50.4]$  and the anomaly disappears.

### C. Eighty-six strategies and the statistics they require

Reporting the best of a large sweep is selection, not discovery. We evaluated 86 parameterised rules, covering momentum and mean reversion at ten lookbacks, six moving average crossover pairs, RSI, Bollinger, breakout and fade variants, MACD, gap, volatility regime, time-of-day, day-of-week and streak rules, on non-overlapping 30-minute holds with 4 bp round-trip cost (Table VII).

The best strategy in the sweep is *worse* than what selection alone typically manufactures. Two methodological notes:

**One-sided tests.** An earlier version of this analysis used a two-sided  $t$ -test and reported “42 of 86 significant.” Those were strategies that reliably *lose*. The hypothesis is profitability, so the test is one-sided; reliable losers are the cost of trading noise, not discoveries.

**Where the standard corrections fail.** The De-flated Sharpe Ratio [16] estimates its null from the cross-sectional variance of trial Sharpes, assuming a common-variance normal sample. Our library violates this badly: strategies differ by an order of magnitude in turnover, so their Sharpe dispersion is dominated by deterministic cost differences rather than by sampling noise, and the formula returns an implausible null benchmark of 7.77. We therefore replaced it with a circular block bootstrap [7], [9] that resamples forward returns in blocks while holding each strategy’s positions and turnover fixed, destroying the signal–outcome relationship and nothing else. Similarly, PBO via CSCV [17] is *degenerate* here. With nothing significantly profitable in-sample there is no selection to overfit, so we report it as such rather than as a result.

TABLE VI

OUT-OF-SAMPLE DIRECTIONAL ACCURACY AT THE 30-MINUTE HORIZON, SCORED THREE WAYS ON IDENTICAL PURGED WALK-FORWARD SPLITS. *PER-BAR* IS WHAT A DASHBOARD REPORTS: ONE FORECAST PER BAR OVER A 6-BAR HORIZON, WITH THE EXACT BINOMIAL INTERVAL ITS NOMINAL  $n = 16856$  IMPLIES. *AT  $n_{\text{EFF}}$*  IS THE SAME POINT ESTIMATE WITH THE INTERVAL WIDENED BY THE DESIGN EFFECT OF (2). *PER-EVENT* RE-SCORES ONE OBSERVATION EVERY 6 BARS, SO NO TWO OBSERVATIONS SHARE A BAR. SHUFFLED-LABEL CONTROLS ARE PER-BAR.

| Model                       | BANKNIFTY   |              |                        |            |       | NIFTY 50   |              |                        |            |       |
|-----------------------------|-------------|--------------|------------------------|------------|-------|------------|--------------|------------------------|------------|-------|
|                             | per-bar     | $p$          | CI at $n_{\text{eff}}$ | per-event  | $p$   | per-bar    | $p$          | CI at $n_{\text{eff}}$ | per-event  | $p$   |
| Majority-class baseline     | 49.3        | 0.061        | —                      | —          | —     | 49.8       | 0.655        | —                      | —          | —     |
| Logistic regression         | 50.6        | 0.095        | [48.6, 52.7]           | 50.1       | 0.895 | 49.3       | 0.082        | [47.3, 51.3]           | 49.1       | 0.355 |
| Gradient boosting           | 50.7        | 0.059        | [48.9, 52.6]           | 50.3       | 0.777 | 49.7       | 0.405        | [47.8, 51.6]           | 48.9       | 0.250 |
| LightGBM                    | 51.1        | <b>0.006</b> | [49.2, 52.9]           | 50.2       | 0.865 | 49.8       | 0.666        | [48.0, 51.7]           | 50.3       | 0.748 |
| XGBoost                     | <b>51.3</b> | <b>0.001</b> | [49.5, 53.2]           | 50.4       | 0.692 | 50.1       | 0.817        | [48.2, 52.0]           | 49.8       | 0.836 |
| $k$ -nearest neighbours     | 50.5        | 0.209        | [48.7, 52.3]           | 49.6       | 0.720 | 50.1       | 0.805        | [48.3, 51.9]           | 50.9       | 0.355 |
| LSTM                        | 50.4        | 0.251        | [48.4, 52.5]           | 51.1       | 0.235 | 49.4       | 0.127        | [47.3, 51.6]           | 49.4       | 0.509 |
| Transformer                 | 50.8        | <b>0.028</b> | [48.9, 52.8]           | 50.9       | 0.355 | 48.3       | <b>0.000</b> | [46.3, 50.4]           | 48.4       | 0.101 |
| Temporal Fusion Transformer | 49.4        | 0.101        | [47.2, 51.5]           | 49.0       | 0.299 | 50.1       | 0.723        | [48.0, 52.2]           | 51.2       | 0.193 |
| Significant at $p < 0.05$   | <b>3/8</b>  |              | <b>0/8</b>             | <b>0/8</b> |       | <b>1/8</b> |              | <b>0/8</b>             | <b>0/8</b> |       |

$n_{\text{eff}} \approx 2100\text{--}3000$  against a nominal 16856; per-event  $n = 2810$ . Shuffled-label controls span 49.5–50.3% on every model and both indices.

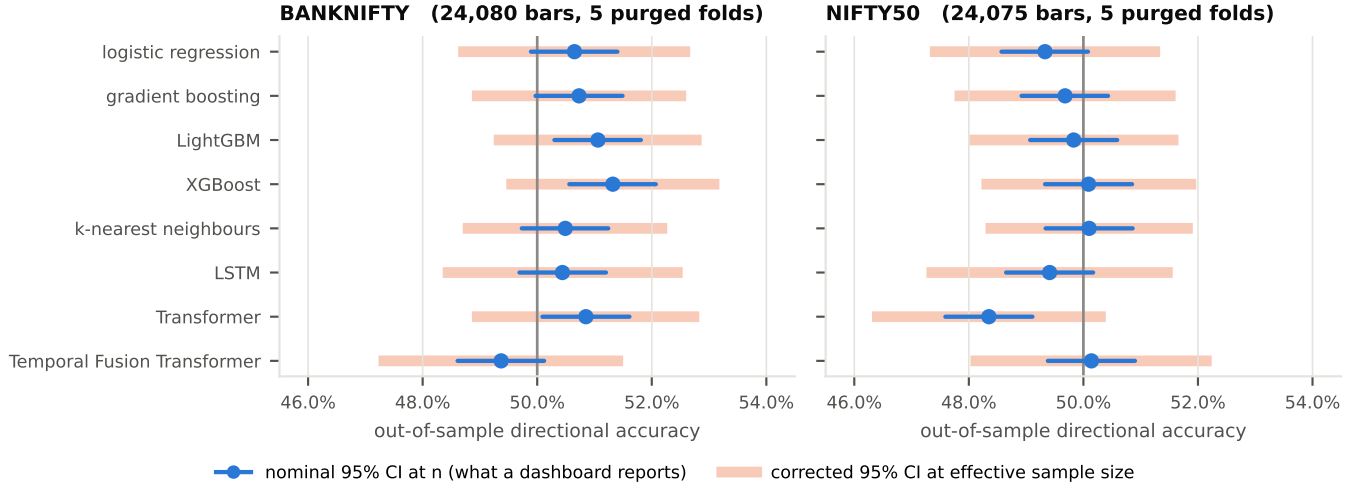


Fig. 3. Nine model families at the 30-minute horizon. The narrow blue interval is the exact binomial interval a dashboard would report at the nominal  $n = 16856$ ; the wide orange band is the same estimate at its effective sample size. Every corrected interval contains 50%. The spread between architectures is smaller than the width of a single corrected interval, which is the sense in which the choice of architecture does not matter here.

TABLE VII

STRATEGY SWEEP WITH MULTIPLICITY CONTROL, BANKNIFTY, 86 TRIALS, NON-OVERLAPPING 30-MINUTE HOLDS AT 4 BP COST.

| Quantity                              | Value        |
|---------------------------------------|--------------|
| Strategies evaluated                  | 86           |
| Positive Sharpe at all                | 2            |
| Significantly profitable (one-sided)  | 0            |
| Surviving Bonferroni                  | 0            |
| Surviving Benjamini–Hochberg, FDR 5%  | 0            |
| Positive and stable in both halves    | 1            |
| Best observed Sharpe                  | 0.21         |
| Bootstrap null, best-of-86, median    | 0.44         |
| Bootstrap null, best-of-86, 95th pct. | 1.31         |
| $P(\text{null} \geq \text{observed})$ | <b>0.658</b> |

Reporting a degenerate statistic as a finding would be an

instance of the error this paper is about.

## VIII. ECONOMIC SIGNIFICANCE

Directional accuracy is the field’s headline metric and is, alone, close to meaningless. Whether a given accuracy earns money depends on the size of the move relative to the cost of capturing it.

For a symmetric bracket with target and stop both at the typical move  $m$  and round-trip cost  $c$ , expectancy per trade is

$$\mathbb{E} = p(m - c) - (1 - p)(m + c) = m(2p - 1) - c,$$

so break-even requires

$$p^* = \frac{1}{2} + \frac{c}{2m}. \quad (3)$$

The ratio  $c/m$  is therefore the entire problem. Table VIII and Fig. 4 evaluate (3) on measured move distributions.

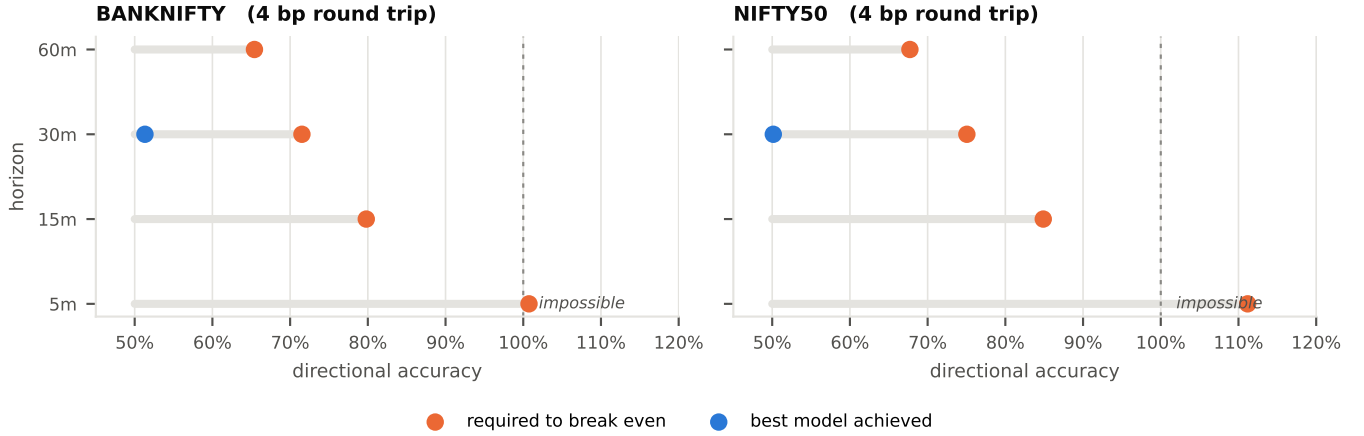


Fig. 4. Break-even accuracy against the best accuracy any model family reached, by horizon. Points beyond the dashed line at 100% are horizons at which no directional accuracy is sufficient.

TABLE VIII  
BREAK-EVEN ACCURACY FROM MEASURED MOVE DISTRIBUTIONS AT 4 BP ROUND-TRIP COST. “ACHIEVED” IS THE BEST OF THE NINE MODEL FAMILIES, SCORED PER EVENT.

| Horizon          | Med. move | $c/m$ | $p^*$          | Achieved      |
|------------------|-----------|-------|----------------|---------------|
| <i>BANKNIFTY</i> |           |       |                |               |
| 5 min            | 22.1      | 1.01  | <b>100.7 %</b> | —             |
| 15 min           | 37.7      | 0.60  | 79.8 %         | —             |
| 30 min           | 52.3      | 0.43  | <b>71.5 %</b>  | <b>50.4 %</b> |
| 60 min           | 72.9      | 0.31  | 65.4 %         | 51.2 %        |
| <i>NIFTY 50</i>  |           |       |                |               |
| 5 min            | 8.1       | 1.22  | <b>111.2 %</b> | —             |
| 15 min           | 14.1      | 0.70  | 84.9 %         | —             |
| 30 min           | 19.7      | 0.50  | <b>75.1 %</b>  | <b>49.8 %</b> |
| 60 min           | 27.7      | 0.35  | 67.7 %         | —             |

Median move in index points.

Two consequences follow immediately.

**At the shortest horizon the requirement exceeds 100% on both indices.** The median five-minute move is smaller than the round trip. There exists no accuracy, whether 90% or 100%, at which five-minute signal trading on these instruments is profitable at retail cost. The shorter the horizon, the more certainly it fails, which inverts the usual marketing of high-frequency retail strategies.

**Instrument choice can make the requirement unreachable.** Table IX evaluates (3) across cost regimes.

“Impossible” denotes  $c > m$ , hence  $p^* > 1$ : the round trip exceeds the typical move and a perfect forecaster loses on the median trade. The instrument most retail participants actually use, over the horizon they actually hold, has no directional accuracy that makes it profitable.

#### A. Statistical significance is not economic significance

Section VII reported an apparent BANKNIFTY effect at  $p = 0.001$  and then removed it under correct scoring. It is worth noting that the effect was economically irrelevant

TABLE IX  
BREAK-EVEN ACCURACY BY COST REGIME, BANKNIFTY 30-MINUTE HORIZON (MEDIAN MOVE 9.3 BP).

| Cost  | Instrument                         | $p^*$             |
|-------|------------------------------------|-------------------|
| 1 bp  | Exchange fees only, no slippage    | 55.4 %            |
| 2 bp  | Tight futures, minimal slippage    | 60.8 %            |
| 4 bp  | Realistic futures round trip       | 71.5 %            |
| 8 bp  | Futures with adverse slippage      | 93.0 %            |
| 20 bp | ATM options: spread + 30 min theta | <b>impossible</b> |
| 40 bp | OTM options or a wide book         | <b>impossible</b> |

even while it stood: 51.3% against a 71.5% requirement loses on every trade, so the correction changed the statistical verdict without changing the practical one.

That is the more general point. The gap between achievable and required accuracy is roughly twenty percentage points, which is far larger than any effect the scoring dispute could move. Fourteen months of accuracy optimisation produced no tradeable result, and would not have produced one even if the apparent edge had been real.

#### B. Reframing high-accuracy claims

Suppose a claimed 85% accuracy at the 30-minute horizon were genuine. It would clear break-even by 13.5 points on BANKNIFTY futures and 9.9 on NIFTY 50, and would remain unprofitable on options. A claim of 85% is therefore either an artefact of the kind measured in Section V, where a zero-skill system reaches that figure occasionally, or it is true and far less valuable than it sounds. Both readings argue against paying for it.

#### C. A filter that worked and still lost

A final corroboration. We trained a meta-label filter [18] in which a secondary model decides whether a primary signal’s setup is worth taking, labelled on *points per minute* rather than on target-hit. The label choice matters:

TABLE X

VOLATILITY PREDICTION THROUGH THE HARNESS OF SECTION IV, 30-MINUTE HORIZON, FIVE PURGED FOLDS. INTERVALS ARE AT THE EFFECTIVE SAMPLE SIZE.

|                                             | per bar | per event | CI at $n_{\text{eff}}$ |
|---------------------------------------------|---------|-----------|------------------------|
| <i>BANKNIFTY</i> ( $n_{\text{eff}} = 976$ ) |         |           |                        |
| LightGBM                                    | 70.8    | 71.3      | [68.0, 73.7]           |
| Persistence baseline                        | 67.4    | 68.0      | [64.5, 70.4]           |
| Shuffled-label control                      | 54.1    | —         | —                      |
| <i>NIFTY 50</i> ( $n_{\text{eff}} = 910$ )  |         |           |                        |
| LightGBM                                    | 72.0    | 72.1      | [69.1, 74.9]           |
| Persistence baseline                        | 67.4    | 68.5      | [64.4, 70.5]           |
| Shuffled-label control                      | 52.2    | —         | —                      |

training on target-hit raises the hit-rate from 42 % to 53 % while making the money worse, because the model learns to prefer small, easy targets.

Out-of-sample over 106 unseen days on BANKNIFTY, the filter cut trades from 2.3 to 1.0 per day and raised quality from +1.44 to +1.82 points per minute, holding in both halves of the test. The total moved from −3777 to −981. A working filter applied to a system with no edge converts a large loss into a small one. It cannot manufacture a profit that is not in the underlying signal.

#### IX. WHAT IS PREDICTABLE INSTEAD

The fifteen production features are all transforms of one price series; reshaping price cannot reveal what price does not contain. What follows comes from changing the target or the information source.

These results have been through the harness of Section IV, and the outcome is the best argument for having built it: the correction that destroyed our directional result in Section VII leaves the volatility result standing.

##### A. Volatility

The target is binary magnitude: will realised volatility over the next 30 minutes exceed the median of the fold’s own training window? The label is thresholded inside each fold, since thresholding on the full sample would leak the test period’s volatility level. Volatility clusters, so the decisive comparison is not against a coinflip but against *persistence*, a rule that predicts BIG whenever the last 30 minutes were above the same threshold (Table X).

Three things separate this from the directional case.

**It survives per-event scoring.** Accuracy moves from 70.8 % to 71.3 % on BANKNIFTY and 72.0 % to 72.1 % on NIFTY 50 when re-scored one observation per horizon. Direction fell from 51.3 % to 50.4 % under the identical operation. The correction is therefore not an instrument that flattens everything it touches; it distinguishes a real effect from an artefact, which is the only reason to trust what it did in Section VII.

**The shuffled control separates cleanly.** At 52–54 % against 71–72 % on real labels, the gap is roughly twenty points. In the directional experiment the same gap was under two.

TABLE XI

SHORT ATM STRADDLE, HELD TO EXPIRY, NET OF A 10 % ROUND-TRIP COST ON PREMIUM. DTE IS THE ENTRY POINT IN TRADING DAYS BEFORE EXPIRY.

|                  | Trades | Win  | Net/trade | Worst | h1/h2 |
|------------------|--------|------|-----------|-------|-------|
| NIFTY, 3 DTE     | 108    | 61 % | +2.8      | −687  | 61/63 |
| NIFTY, 1 DTE     | 108    | 62 % | +3.2      | −439  | 61/72 |
| BANKNIFTY, 3 DTE | 39     | 64 % | +14.2     | −881  | 58/75 |
| BANKNIFTY, 1 DTE | 40     | 68 % | +59.3     | −936  | 75/70 |

Net per trade and worst trade in index points.

#### But the edge over persistence is not established.

The model beats “the next thirty minutes look like the last thirty” by only +3.3 points on BANKNIFTY and +3.7 on NIFTY 50 per event, and at  $n_{\text{eff}}$  the model and baseline intervals overlap. Most of a 71 % headline is volatility clustering, which has been documented since Engle [22], [23] and needs no model. What the model adds is real but small, and we do not claim it is significant.

The useful statement is therefore narrower than “72 % accuracy”. It is that magnitude carries far more structure than direction on identical features, and that most of that structure is already in the previous half hour.

##### B. The variance risk premium

Short at-the-money straddles entered three trading days before expiry and held to expiry, priced on real end-of-day premiums from the F&O archive, with a round-trip cost of 10 % of premium collected (Table XI).

The premium sold exceeds the subsequent move on average in every configuration, which is the variance risk premium doing what the literature says it does [25], [26]. It is structural rather than predictive, which is why it survives a protocol that eliminated the predictive results.

It is also the weakest-evidenced claim in this paper, and four caveats are required. The sample is 39–108 expiries, not thousands. No dispersion statistic accompanies the mean, so the net figures carry no interval. No multiplicity correction is applied over the choice of strike, entry day and holding rule, and Table XI reports four such choices. And BANKNIFTY’s 39 expiries against NIFTY’s 108 reflect NSE moving BANKNIFTY options from weekly to monthly expiry partway through the sample, so the two rows are not a like-for-like comparison and the BANKNIFTY figures cover a shorter, differently structured period.

We note also that an earlier version of this work reported +22.5 points per trade for NIFTY and a *loss* of 7.3 points for BANKNIFTY. Neither reproduces from the committed script; the correct figures are above. That error is precisely the failure mode this paper argues against, and it survived until the analysis was re-run rather than remembered.

Most importantly, the payoff is short-volatility: a 61–68 % win rate on small credits against occasional losses of 400–900 points. A high win rate here is a statement about

tail exposure, not about skill, and the structure must be implemented with defined risk.

### C. Option-chain positioning

### D. Edges decay

The single cost-positive signal we found was a contrarian rule on put–call ratio extremes: 204 trades, profit factor 1.11 over two years. It was already dead when we discovered it. Directional accuracy of 74 % in 2024 fell to 36 %, with a profit factor of 0.74, over the final six months. This is the mechanism Timmermann and Granger describe [21], whereby discovered patterns self-destruct. It also bears on sustained-accuracy claims: a genuinely real edge in this data died inside two years, so a claim of stable accuracy over 76 trading days carries no information even if every number in it is true.

## X. LIMITATIONS

We state these plainly because the paper’s argument is about honest reporting.

Two indices carry the primary intraday analysis, over the seventeen months of 5-minute history available to us, in one market. The break-even model (3) assumes a symmetric bracket; asymmetric payoffs lower the required hit-rate while also lowering the achieved one, and we did not sweep that trade-off. Our Temporal Fusion Transformer is a compact variant retaining variable selection, gated residual networks and interpretable attention, but omitting static covariates and the quantile head, and should not be read as the published architecture [33]. Some results in our research record come from scripts not retained; these are labelled in the repository and none are load-bearing for Sections V–VIII. Index spot carries no volume, so any VWAP-derived feature is misnamed, and whether a true volume-weighted average helps is untested. Finally, the order-flow hypothesis, where the literature locates genuine short-horizon alpha [27], [28], remains untested here for want of history: we run a live collector, but message-level history is not sold to retail accounts.

## XI. CONCLUSION

The headline metric of intraday directional forecasting is not merely noisy. It is systematically biased upward by the way it is computed, and the bias is large enough to manufacture exactly the figures used in marketing: a forecaster with zero skill posts a  $\geq 71\%$  session once in 51, and once every seven sessions across a modest panel, while reporting a confidence interval  $1.66\times$  too narrow.

Scored correctly, nine model families reach at most 51.3 %, 86 strategies produce a winner worse than a bootstrap null, and the small effect that does survive every control sits twenty percentage points below its own break-even threshold. On the instruments and horizons most retail participants use, that threshold exceeds 100 %.

What remains is not direction. Magnitude is forecastable and, tellingly, unharmed by the same correction

that removed the directional effect, though most of its apparent accuracy is volatility clustering that a persistence rule already captures. The variance risk premium is real and structural. Order-flow data, which we could not obtain, is where the literature locates the remaining short-horizon alpha.

Reporting practice should change to match, and the change is cheap. Section VI gives an estimator whose interval covers at 94 % where the conventional one covers at 40 %, at no cost to the point estimate. Scoring unit, effective sample size, multiple-testing correction, a shuffled-label control and an explicit cost model are not optional refinements to an accuracy figure. Without them the figure does not mean anything.

## REPRODUCIBILITY

Code, the experiment harness, raw results, figure generation and the full research record, including five formally retracted claims, are available at <https://github.com/rachitranka25/accuracy-illusion>. Experiments use a fixed seed and pin native thread pools to one thread.

Opt plus .5pt Opt plus 0.3pt

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